AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 1 · STUDENT LESSON

Predictive and Generative AI

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Before you start

AI-2 Week 27: identify retrieval, prediction and generation within one pipeline.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Distinguish discriminative prediction from generation by inspecting the requested output. Try the recall before opening notes.

Your learning target

I can distinguish discriminative **prediction** from generation by inspecting the requested **output**.

I can calculate an empirical next-token distribution from counted continuations.

I can separate next-token fit from factual support and downstream usefulness.

Remember before reading

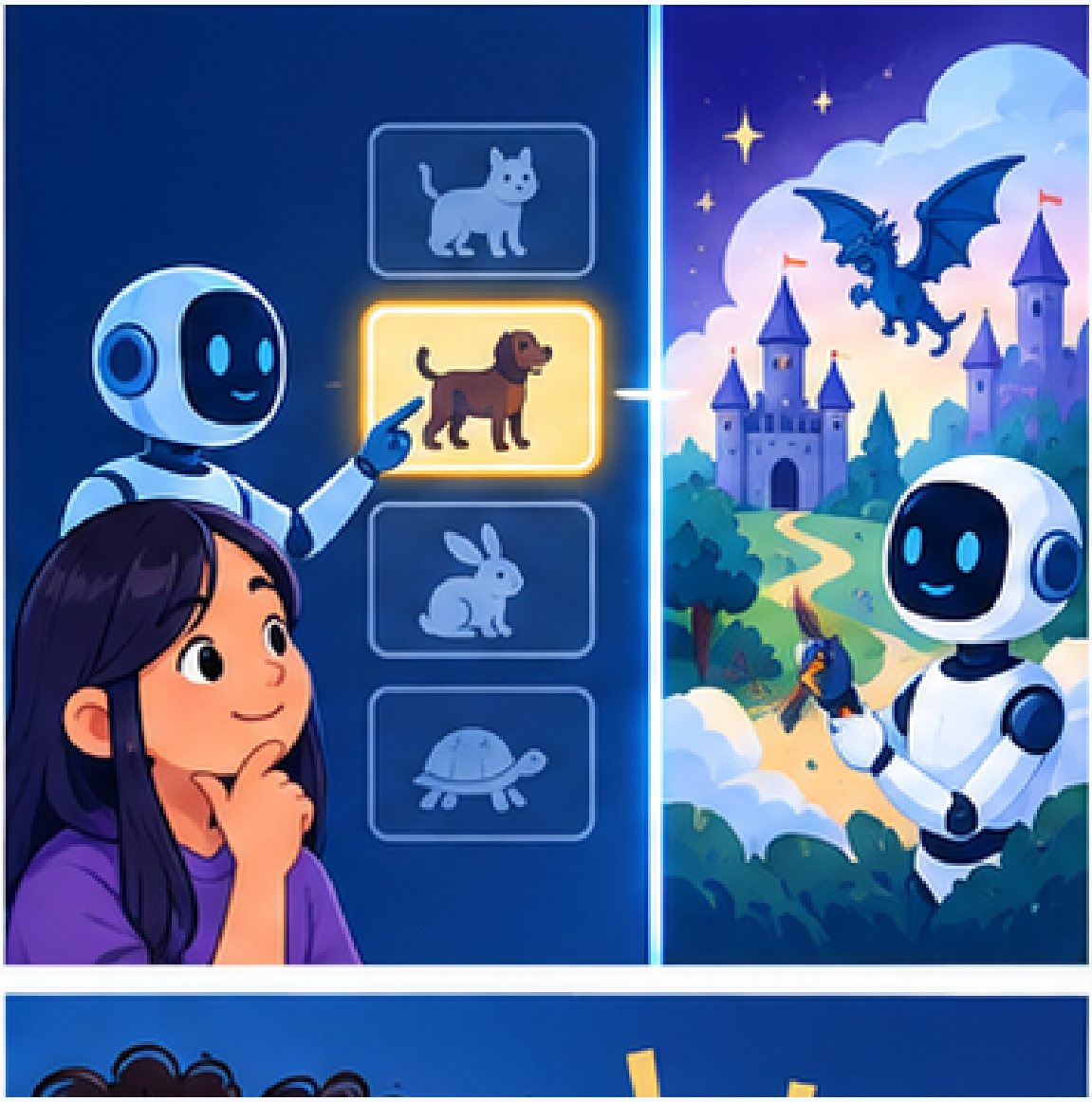
Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Capstone Expo. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 1: Predictive and **Generative AI**

Prediction estimates a **target**; generation samples a new structured **output**.



CREATOR MISSION

Classify systems and explain the **output space** for each.

Creator name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

What is the difference between a model **prediction** and **evidence** that a statement is true?

This week's end goal

By the end, I can identify generation and **prediction** objectives without confusing either with verified truth.

STUDENT LESSON · 2 OF 18

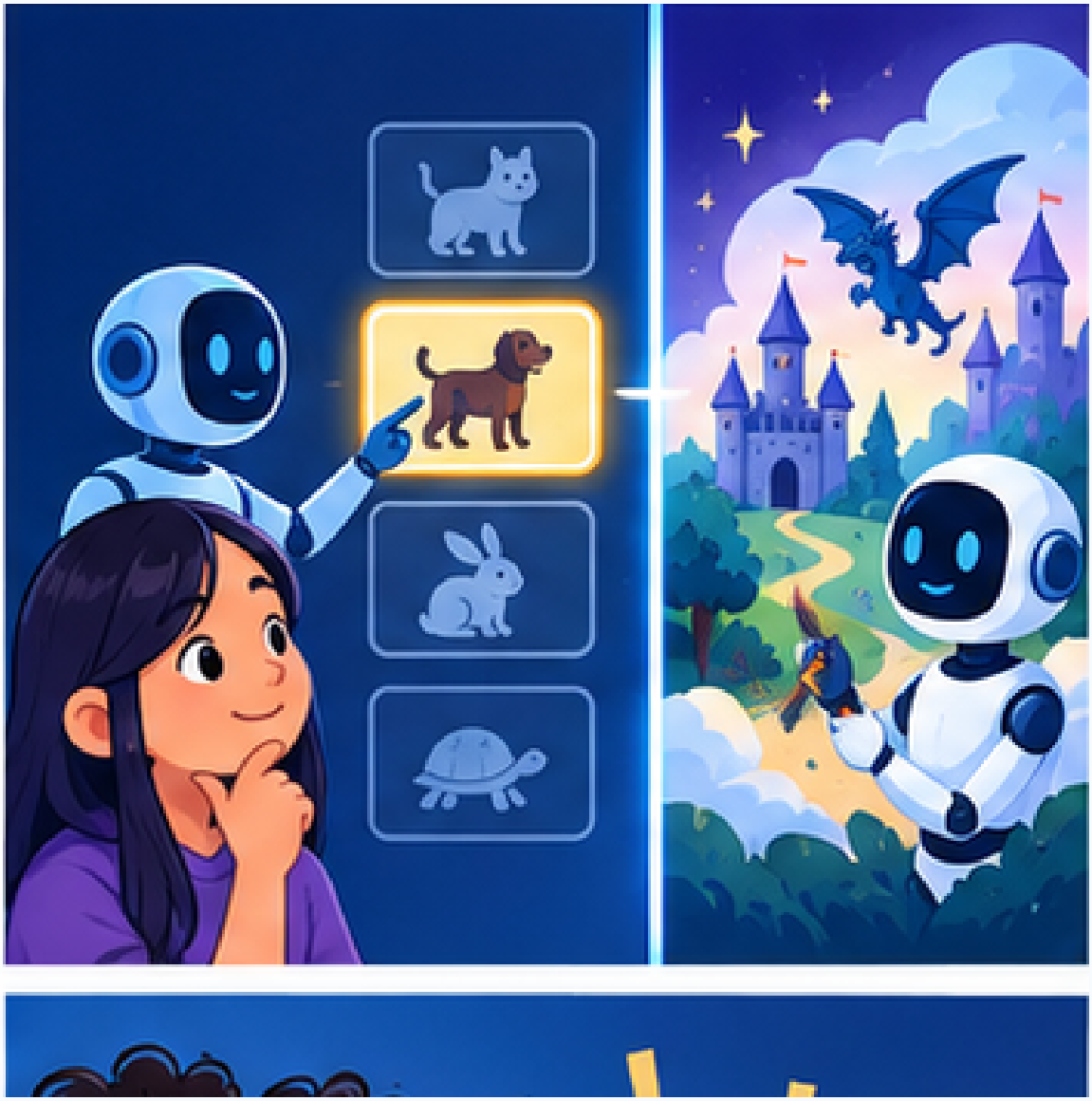
Notice the evidence

Investigate the supplied Predictive and Generative AI case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use predictive and **generative ai** to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

A classifier might map a message to the label request or report. A text generator instead produces a sequence, predicting a distribution over the next token given a prefix.

Both make **prediction**s, but the **training target**s and permitted **output**s differ. One application can combine generation with a classifier, retrieval and human approval.

Calling a whole product “generative” does not describe every component.

The objective defines what **training** attempts to fit. Next-token fit rewards matching observed continuations; it does not directly certify that an answer is true, well sourced or permitted.

A fluent completion is still an **output** to evaluate against the task.

See how it works

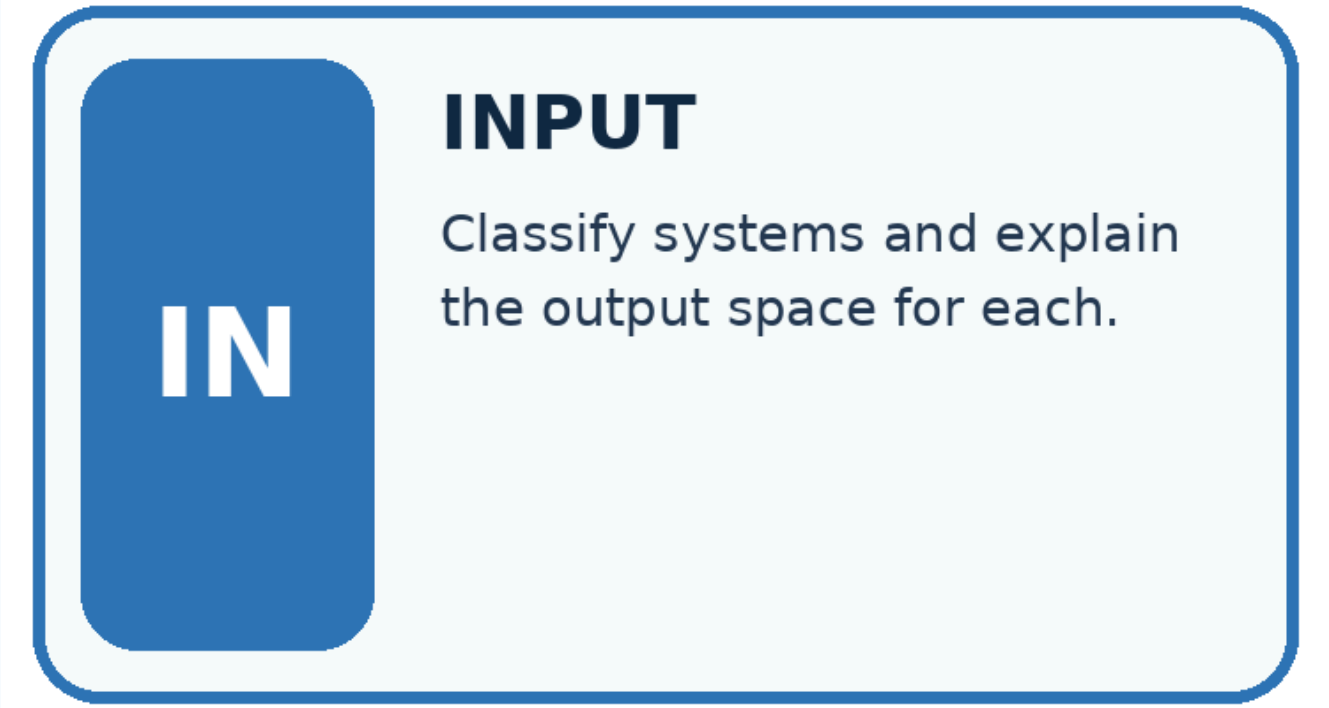
> Follow the mechanism for Predictive and Generative AI. Connect each labelled input, deciding

> step and result.

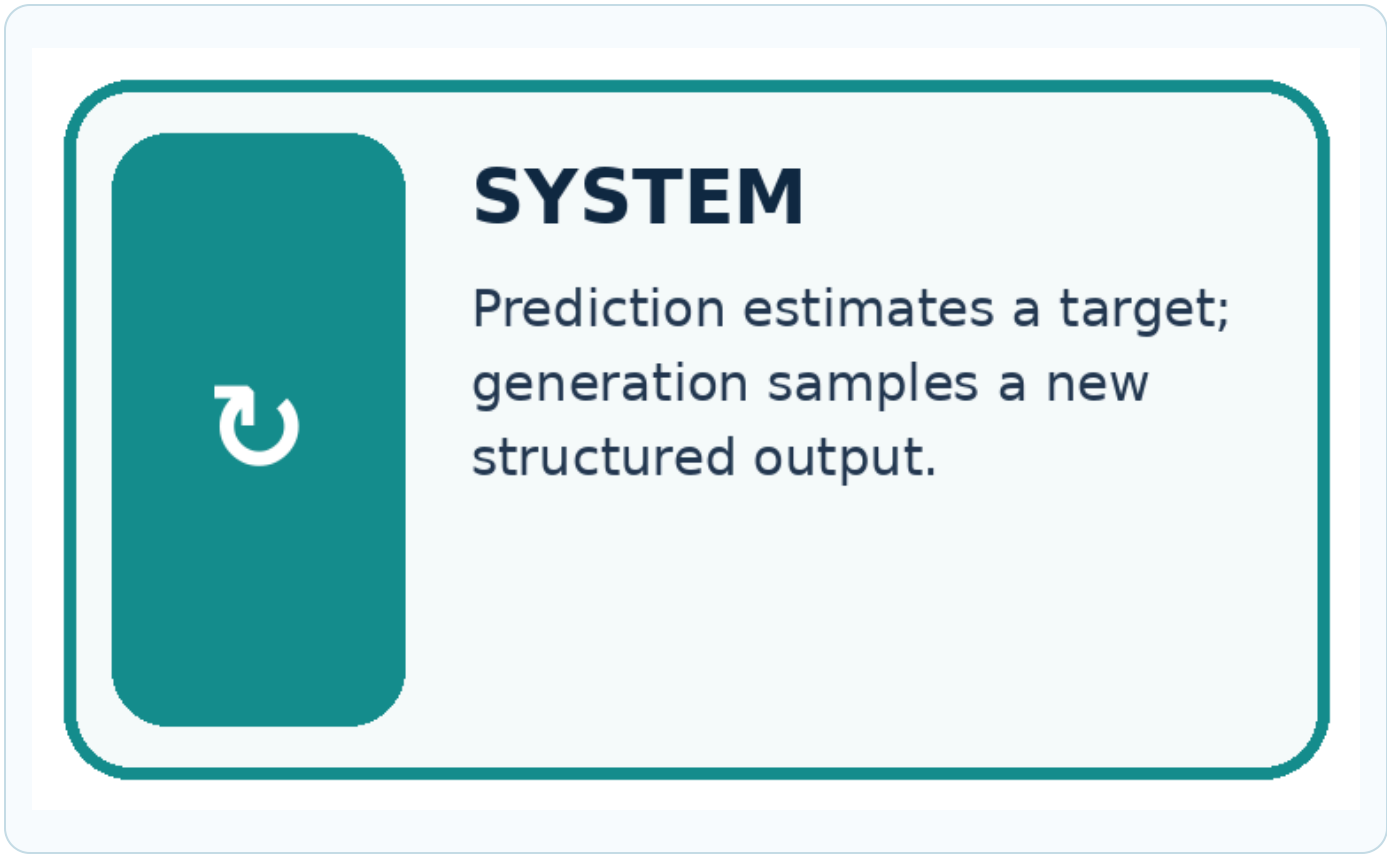
VISUAL MECHANISM

Input → Process → Output

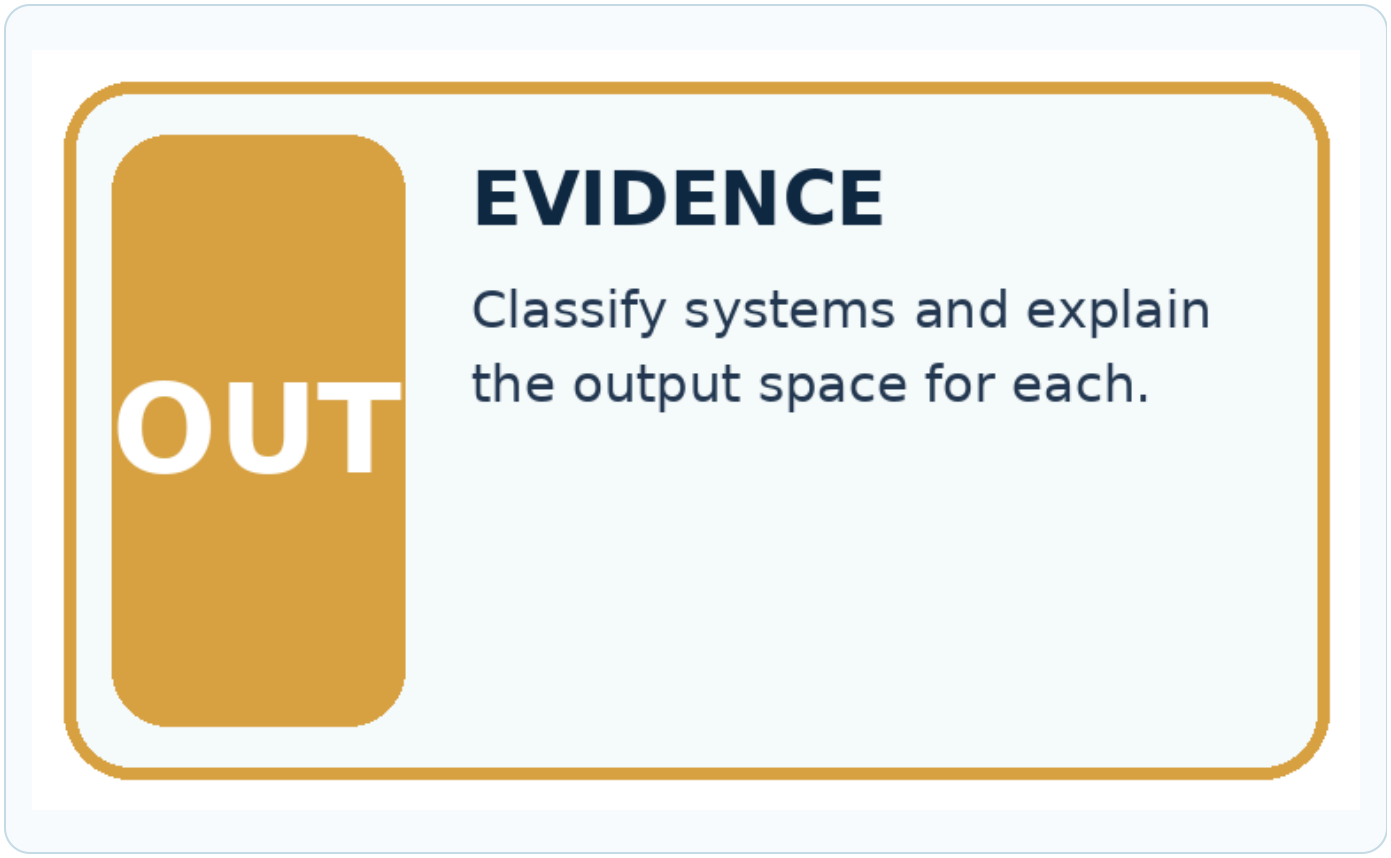
Trace the information. At each stage, ask what changed and how you could test it.



↓


SYSTEM
Prediction estimates a target; generation samples a new structured output.

↓


OUT
EVIDENCE
Classify systems and explain the output space for each.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Prefix	Continuation	Count
The marker is	red	6
The marker is	blue	3
The marker is	green	1

There are ten observed continuations. The empirical probabilities are red 6/10 = 0.6, blue 3/10 = 0.3, green 1/10 = 0.1. They are nonnegative and sum to 1.

A simple frequentist classroom model uses these relative counts as its next-token distribution. These probabilities describe the supplied corpus counts, not the current colour of any actual marker.

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
Ten corpus continuations	Count red 6, blue 3, green 1	Distribution 0.6, 0.3, 0.1
Prefix: The marker is	Select a next token	An output, not an observation
Current marker question	Check direct evidence	Colour claim needs support

The continuation model and a factual observation have different evidence sources.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

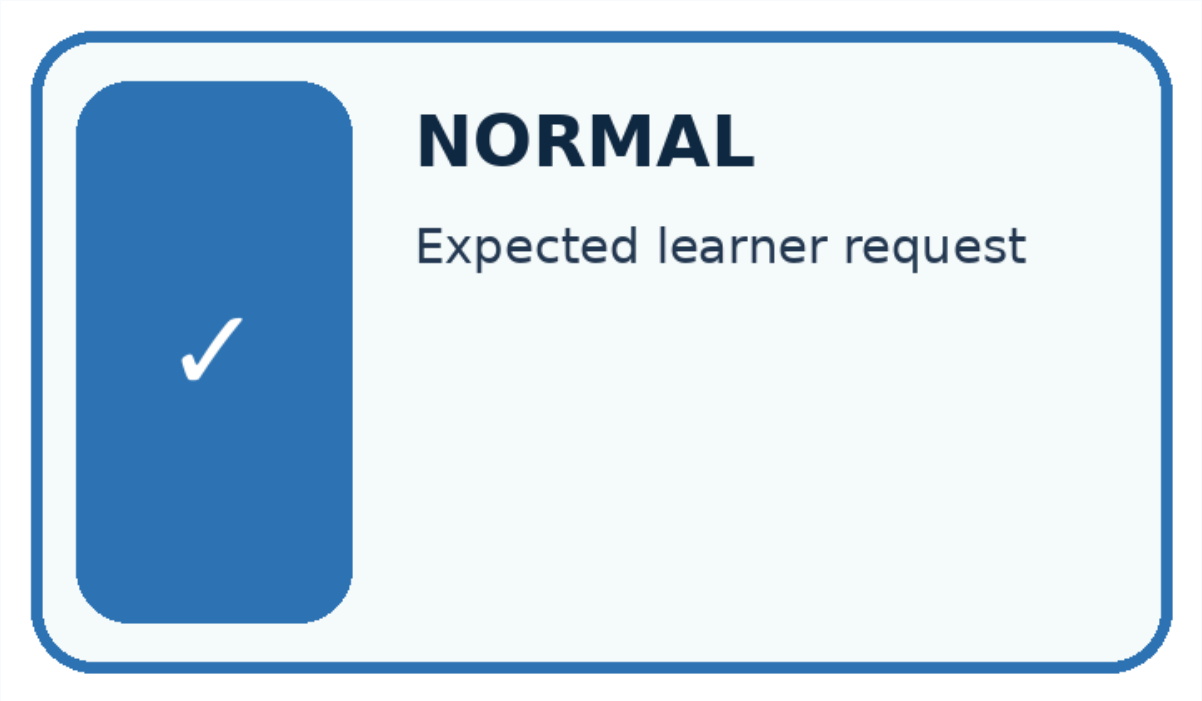
> Compare the supplied cases for Predictive and Generative AI. Identify the detail that changes the conclusion.

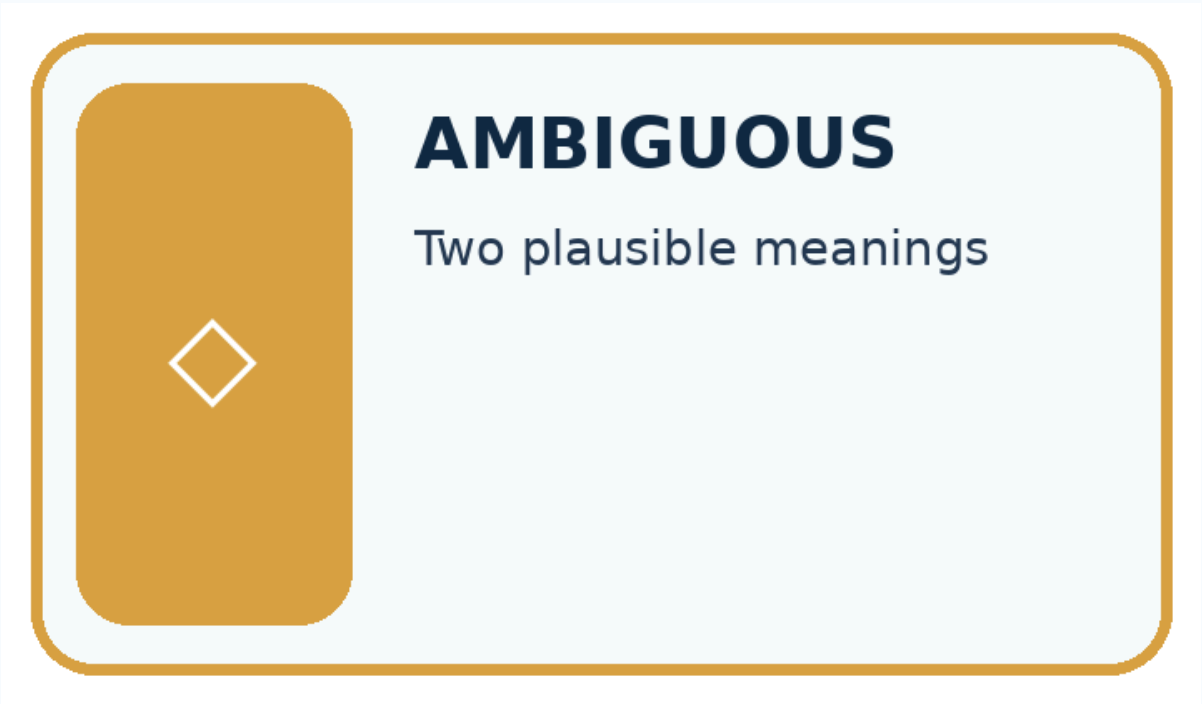
The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

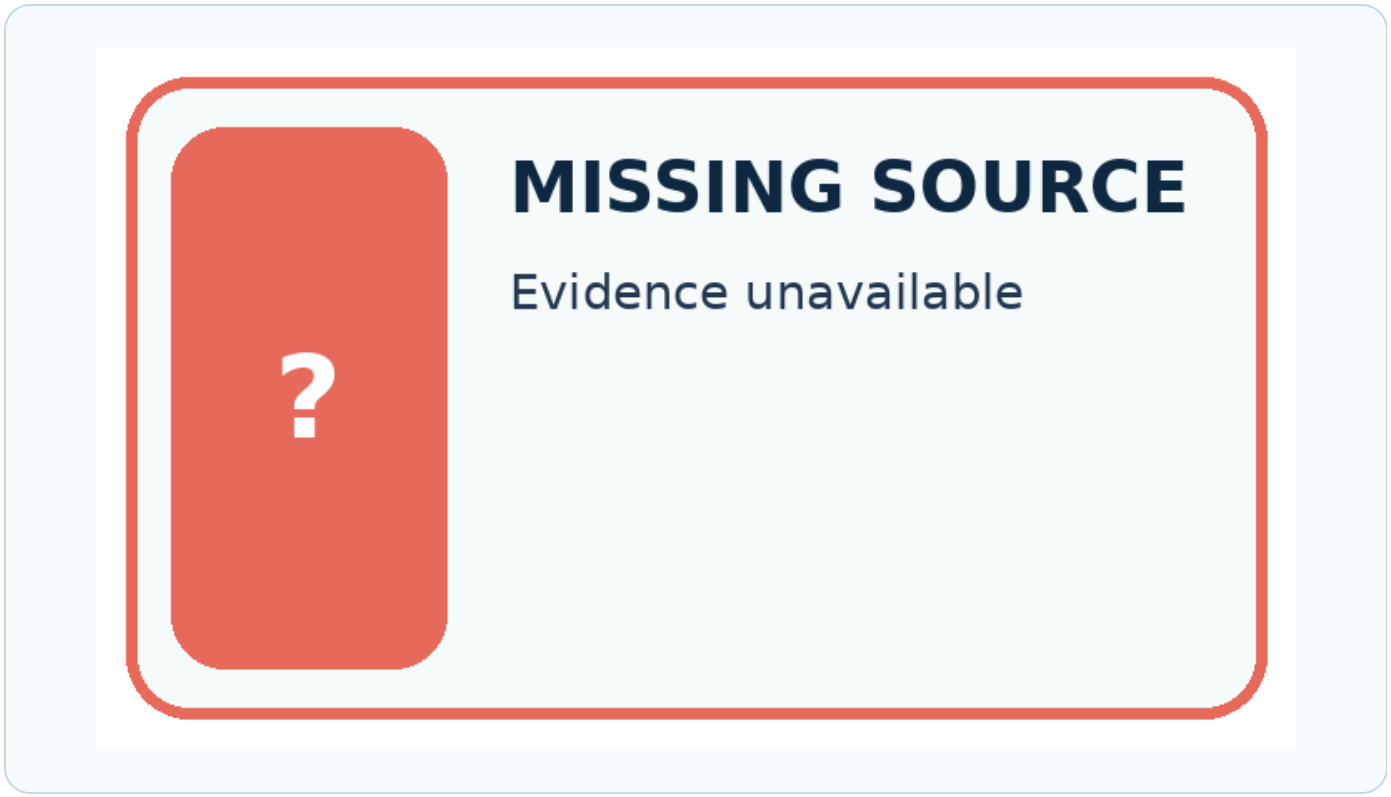
A system is not understood until it survives more than one comfortable example.

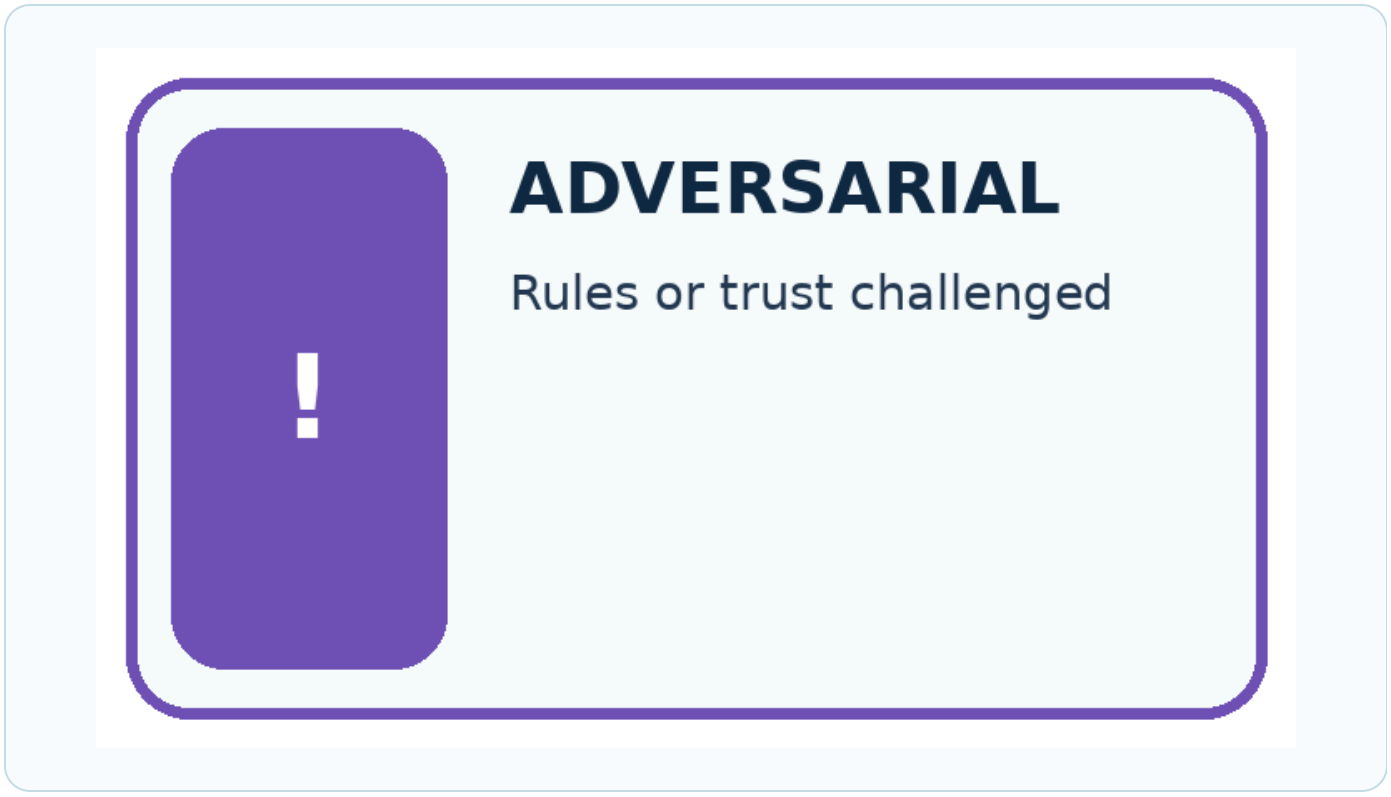




The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.


?
MISSING SOURCE
Evidence unavailable


!
ADVERSARIAL
Rules or trust challenged

Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: The classifier instead predicts history while N and the draft stay unchanged.

Predicted result

Prediction: The category **prediction** changes, but the same unsupported draft claim still needs checking.

Why the result follows

Reason: Stages can fail independently; one result does not validate another.

STUDENT LESSON · 5 OF 18

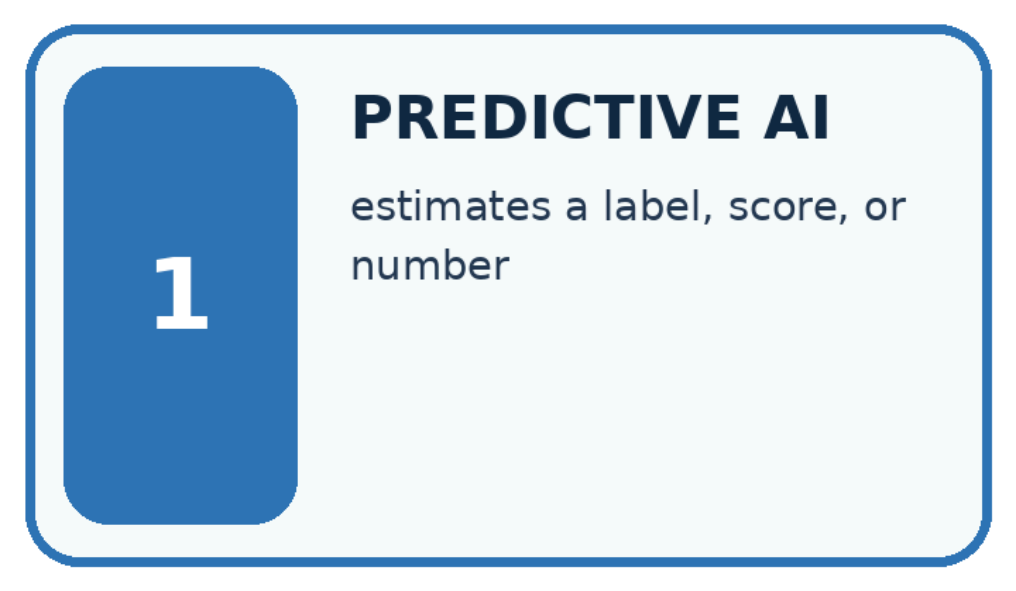
Words that unlock the idea

> Use the vocabulary of Predictive and Generative AI to explain a concrete decision, rather than only memorising definitions.

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

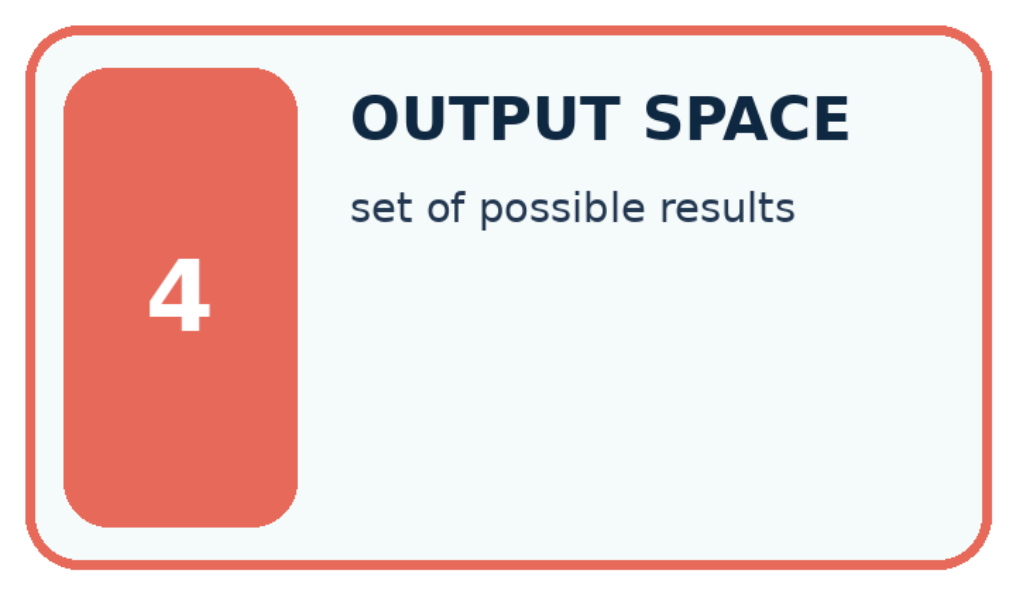
GENERATIVE AI

constructs new content from learned patterns

3

TARGET

value a predictive system estimates



Word	My own example	Not this / boundary
predictive AI	AI used to estimate a target such as a category, number or future outcome from inputs.	
generative AI	AI that produces new content such as text, images or audio rather than only selecting a category.	
target	The category, value or outcome a prediction task is intended to estimate.	
output space	The set or range of outputs a system can produce for a task.	

Words used in the closer explanation

Word	Meaning
Generative objective	A training aim concerned with producing or modelling possible outputs
Discriminative objective	A training aim mapping inputs to a specified target such as a label
Conditional distribution	Probabilities for alternatives given stated information
Autoregressive generation	Producing a sequence one next item at a time using preceding items

STUDENT LESSON · 6 OF 18

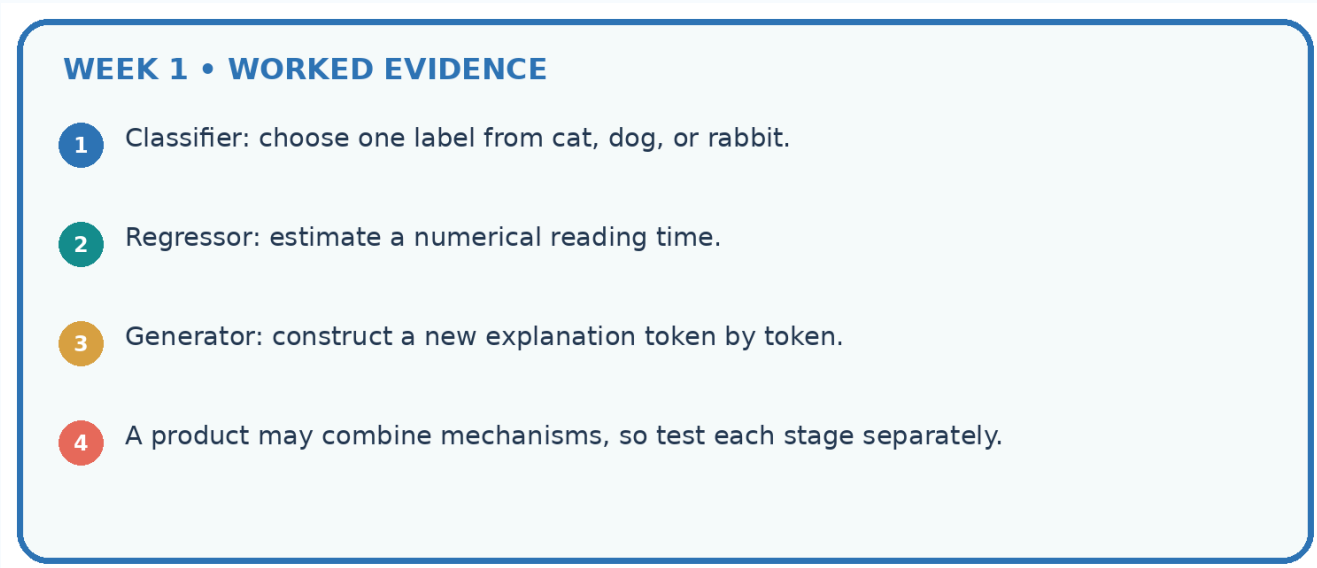
Follow a worked example

> Follow the worked trace for Predictive and Generative AI, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Classifier: choose one label from cat, dog, or rabbit.

2. Regressor: estimate a numerical reading time.

3. Generator: construct a new explanation token by token.

4. A product may combine **mechanism**s, so test each stage separately.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Worked explanation · Predictive and Generative AI

After producing red, the next step conditions on the expanded prefix “The marker is red”. A second table might give an end token probability 0.8 and the token today probability 0.2.

A generated sequence can stop at the end token or continue. Do not reuse the first-step table unchanged after adding a token unless the model explicitly ignores the new prefix.

Worked explanation · Predictive and Generative AI

A separate classifier can return red as a label from an image; that is a different **input** and **target** from continuing text. A downstream evaluation asks whether the system correctly reports the current marker.

It must include ground truth or sources and appropriate failure handling, not just count plausible phrases.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Predictive and Generative AI

A fictional learning tool first classifies a question, then drafts an answer from a supplied note.

The facts we will use

Supplied facts

A fitted classifier labels a question science; its **training** used labelled questions.

Supplied facts

Approved note N: evaporation changes liquid water into water vapour. A generator drafts: “Water evaporates into vapour and always disappears forever.”

AI ACADEMY / AI-3/1

Predictive and Generative AI

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Predict category

The classifier predicts science from the new question.



2. Retrieve note

N supplies a specific statement about evaporation.



3. Generate draft

The draft adds "always disappears forever" beyond N.



4. Verify claims

Keep the supported change-of-state claim; remove or review the unsupported addition.

CHANGE ONE THING

The classifier instead predicts history while N and the draft stay unchanged.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Predict category	The classifier predicts science from the new question.
2. Retrieve note	N supplies a specific statement about evaporation.
3. Generate draft	The draft adds “always disappears forever” beyond N.
4. Verify claims	Keep the supported change-of-state claim; remove or review the unsupported addition.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Classification predicts a category under a trained mapping. Generation produces a sequence of words under a language objective. Supplying a note can support factual grounding, but the draft still requires claim-level checking.

The category being correct does not prove every sentence in the answer is correct.

What this example does not establish

Keep the stated limit

This example does not expose a real model’s architecture or prove that source-fed generation always stays grounded.

Keep the stated limit

Read the labelled picture: Identify the first unsupported claim and explain what source would be needed before asserting it.

Keep the stated limit

Change one thing: The classifier instead predicts history while N and the draft stay unchanged. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Predictive and Generative AI claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: Every AI system that produces an answer is generative. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

Keep three questions separate: did the model fit continuation data; did the **output** satisfy the requested format or task; and are factual claims supported?

A system that drafts a fictional description has different criteria from one reporting a live measurement. Record the source and time when a claim refers to changing conditions.

No AI account is needed to reason about these distinctions.

Exit: a team says “the highest-probability continuation proves the current fact”. Name the evidence they actually have and the evidence still needed.

STUDENT LESSON · 8 OF 18

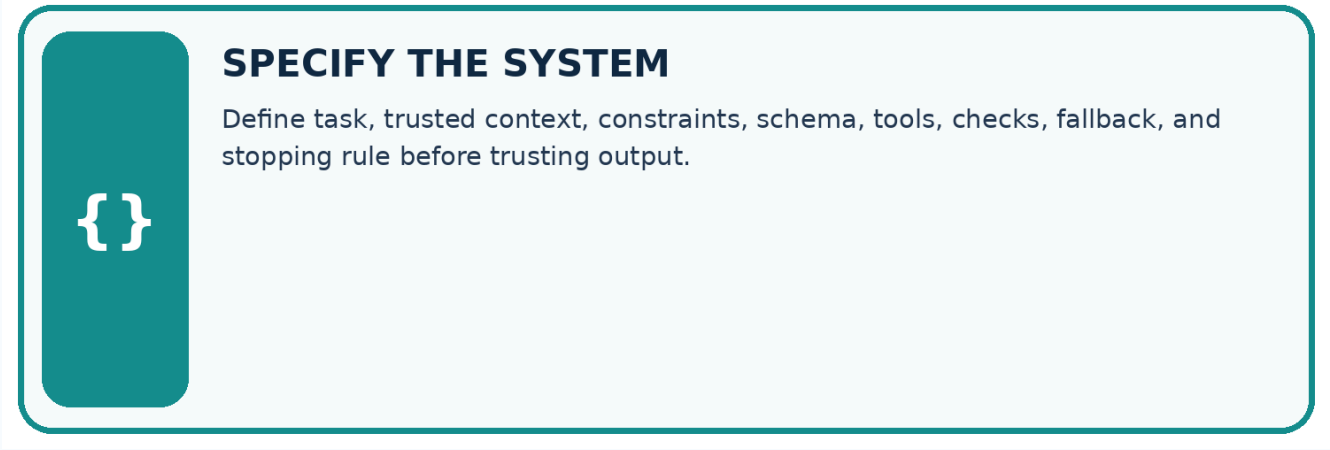
Practise together

Practise Predictive and Generative AI with support, then explain the deciding step in your own words.

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Supported case: the prefix “The door is” is followed in a tiny corpus by open four times and closed once. The empirical probabilities are 4/5 and 1/5. A current sensor reports closed.

Selecting the most frequent continuation open would be plausible under the corpus model but inconsistent with the sensor **observation**.

Guided: compute both probabilities and explain why selecting open does not establish the current door state.

Practise with less help: Compute the distribution

Use workbook W2 for the connected example “Generative and predictive objectives”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Generative and predictive objectives. Proposed explanation: “Generative models do not predict”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Identify the first unsupported claim and explain what source would be needed before asserting it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

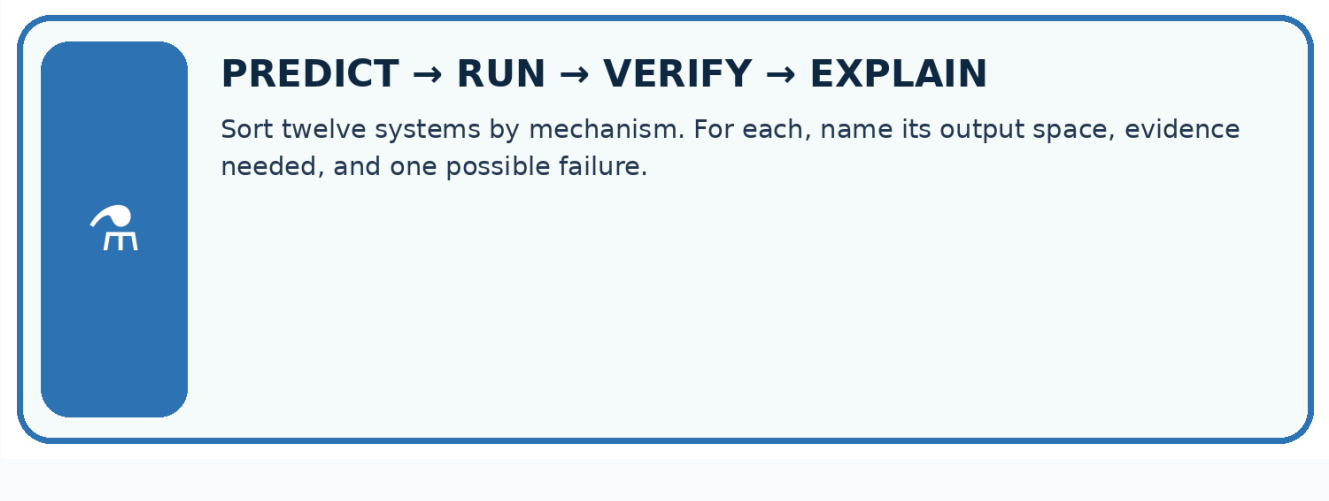
Run the investigation

> Investigate Predictive and Generative AI. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Sort twelve systems by **mechanism**. For each, name its **output space**, **evidence** needed, and one possible failure.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

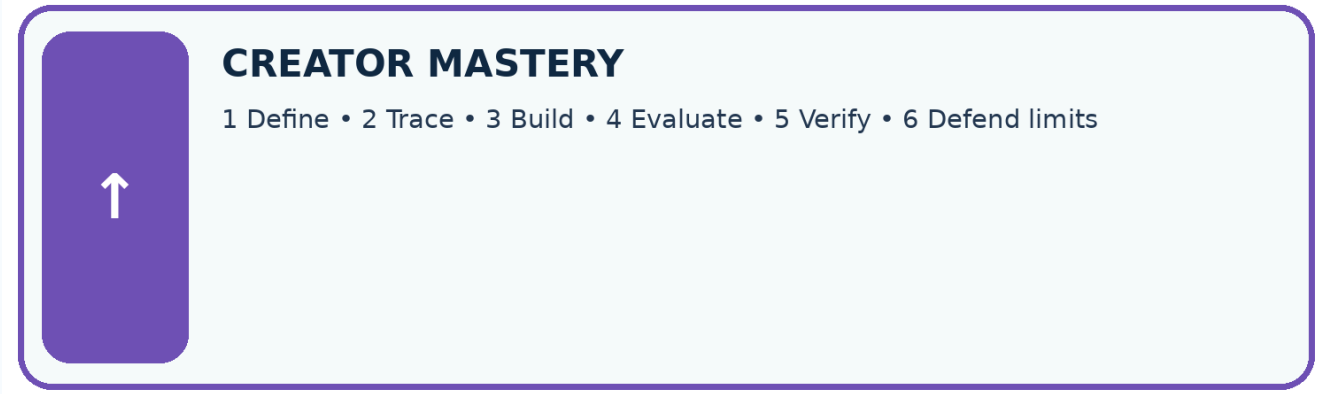
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Predictive and Generative AI. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

> **Write or draw your reasoning and evidence.**

STUDENT LESSON · 11 OF 18

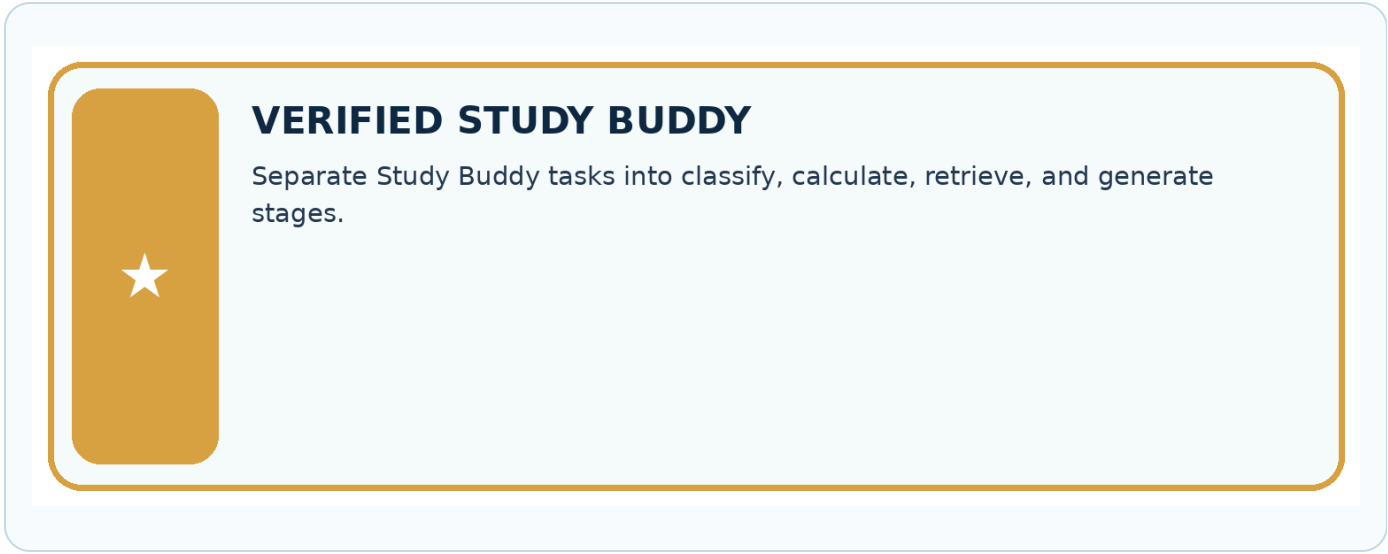
Connect your project

> Apply this week's idea to Verified Study Buddy. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Separate Study Buddy tasks into classify, calculate, retrieve, and generate stages.

Trace the system

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent **output** with verified truth.

Evidence for your project

For the Verified Study Buddy, separate routing **prediction**s, source retrieval and generated explanations in its **evidence** record.

STUDENT LESSON · 12 OF 18

Show what you know

Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.

INDEPENDENT CREATOR PROOF

Classify systems and explain the output space for each.

Try a more demanding case

CHALLENGE Classify systems and explain the **output space** for each.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Predictive and Generative AI case for supported practice. Repeated examples do not become fresh assessments.

Week 1 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A tool labels a question biology, writes a new question and computes 7 times 8.

The question

Identify each operation.

Worked explanation

Labelling selects a category; generation creates text; calculation returns 56. Generated questions need factual and instructional checks.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Three outputs need three different checks

Evidence or stage	Value or result
Classify	Select biology as a category
Generate	Compose a new question
Calculate	7 times 8 equals 56

A tool labels a question biology, writes a new question and computes 7 times 8.

Identify each operation.

Labelling selects a category; generation creates text; calculation returns 56. Generated questions need factual and instructional checks.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Changed independent case

A system retrieves an approved Moon note, uses a trained classifier to predict reading difficulty, then writes a poem. Identify each **mechanism** and the **evidence** needed to check its **output**.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 1 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> A system labels a message as spam. Another writes a new reply. Which task predicts a category and which generates content? Can the new reply contain a false fact?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Separate Study Buddy tasks into classify, calculate, retrieve, and generate stages.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

A study tool finds a paragraph, labels its difficulty and invents practice questions. Identify the three operations and one check for each.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Predictive and Generative AI, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional learning tool first classifies a question, then drafts an answer from a supplied note.

By the end, I can	How I will show it
Distinguish discriminative prediction from generation by inspecting the requested output	Explain the deciding step, show a trace or test, and state one limit.
Calculate an empirical next-token distribution from counted continuations	Explain the deciding step, show a trace or test, and state one limit.
Separate next-token fit from factual support and downstream usefulness	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Predictive and Generative AI without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Distinguish discriminative **prediction** from generation by inspecting the requested **output**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional learning tool first classifies a question, then drafts an answer from a supplied note.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Keep the supported change-of-state claim; remove or review the unsupported addition.

1. Predict category

2. Verify claims

3. Retrieve note

4. Generate draft

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Verify claims Keep the supported change-of-state claim; remove or review the unsupported addition. Classification predicts a category under a trained mapping.

Generation produces a sequence of words under a language objective. Supplying a note can support factual grounding, but the draft still requires claim-level checking.

The category being correct does not prove every sentence in the answer is correct.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The classifier instead predicts history while N and the draft stay unchanged. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. The category **prediction** changes, but the same unsupported draft claim still needs checking.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The category **prediction** changes, but the same unsupported draft claim still needs checking. Stages can fail independently; one result does not validate another.

Repair a missing building block

Review Capstone Expo (ai-2/36). Goal: Defend an application using reproducible **evidence**. Short practice: Give the original winning ID, its score, and counts of eligible, excluded and invalid rows..

Return to this week and explain what you changed.

Keep your completed work

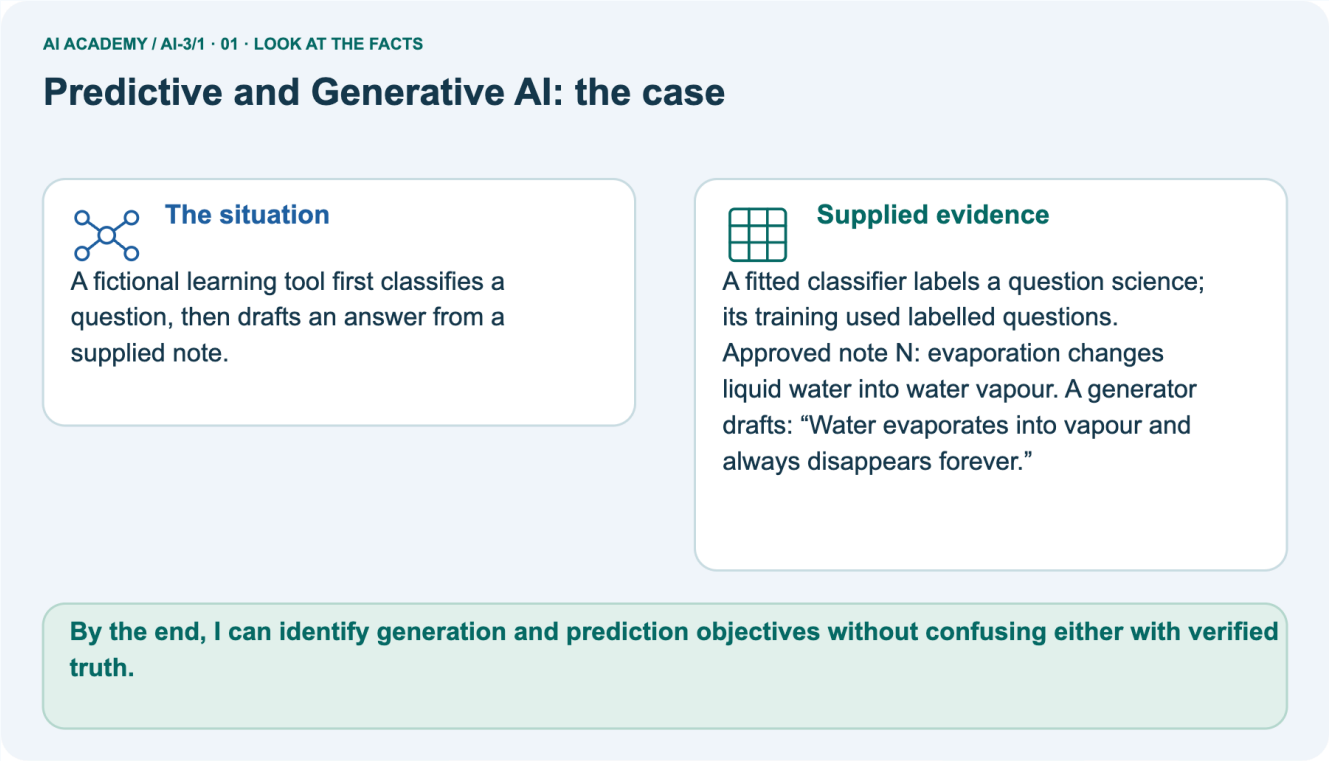
Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

Visual concept journey

> Explain each Predictive and Generative AI graphic using its labels, deciding evidence and relevant limitation.

01. Predictive and Generative AI: the case



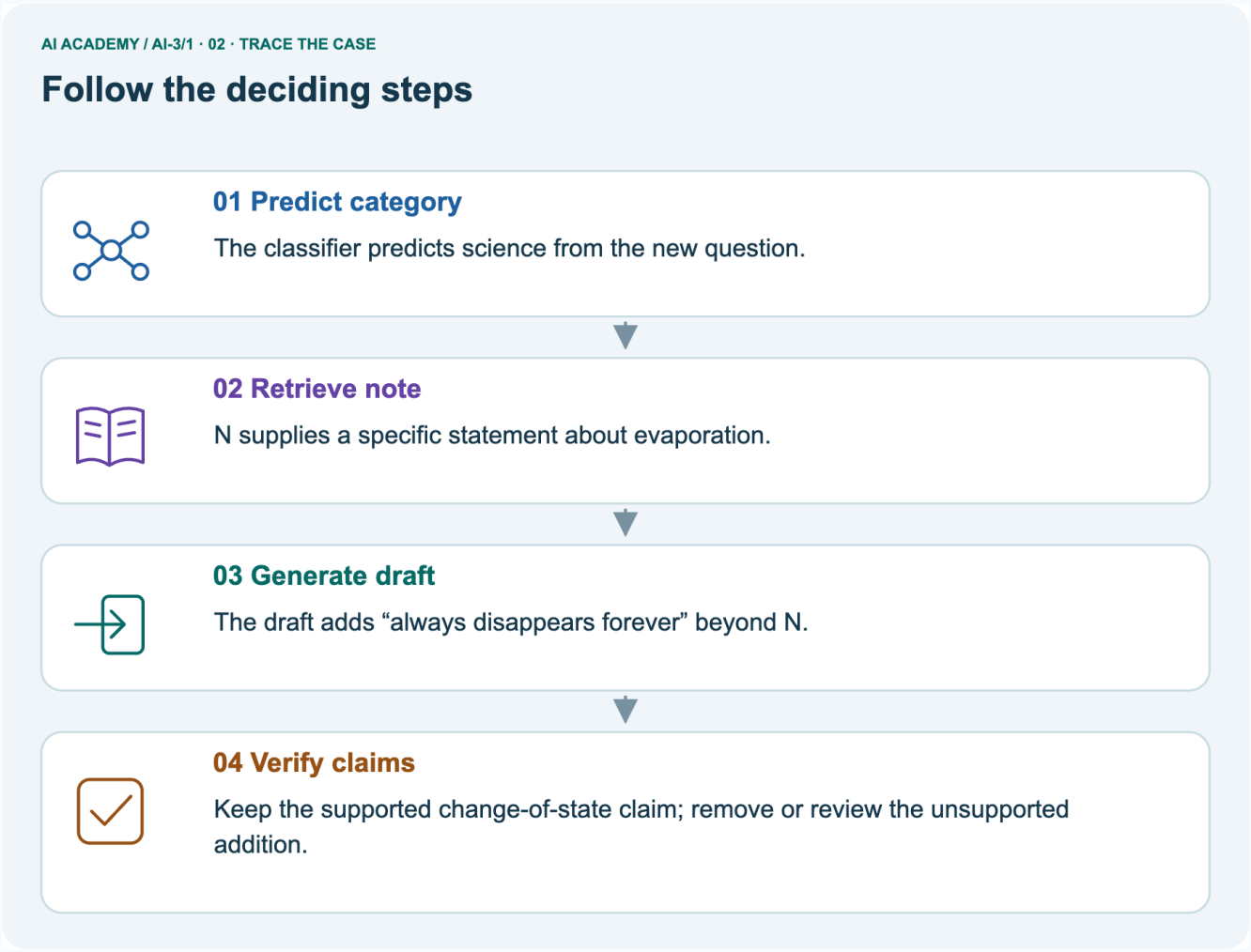
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional learning tool first classifies a question, then drafts an answer from a supplied note. Supplied facts: A fitted classifier labels a question science; its **training** used labelled questions.

Approved note N: evaporation changes liquid water into water vapour. A generator drafts: “Water evaporates into vapour and always disappears forever.”

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

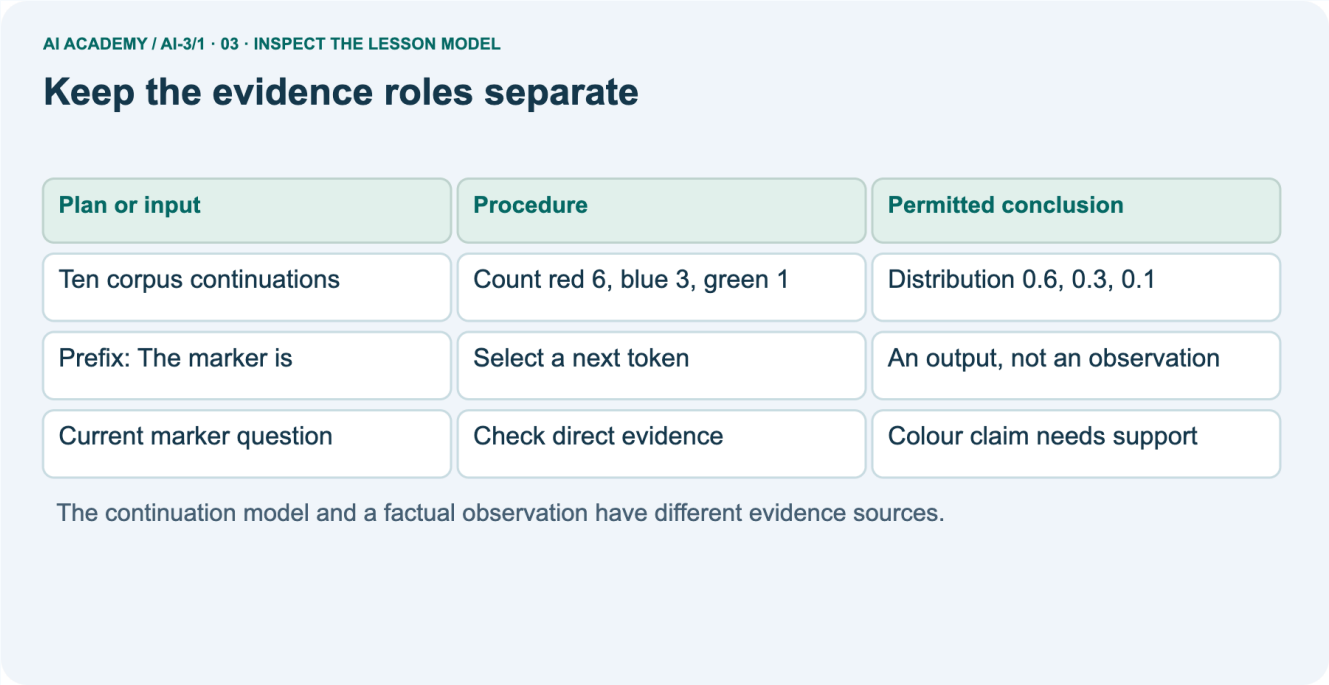


Follow the labelled picture with the matching explanation. Open the vector diagram

Predict category: The classifier predicts science from the new question. → Retrieve note: N supplies a specific statement about evaporation. → Generate draft: The draft adds “always disappears forever” beyond N. → Verify claims: Keep the supported change-of-state claim; remove or review the unsupported addition.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Keep the evidence roles separate



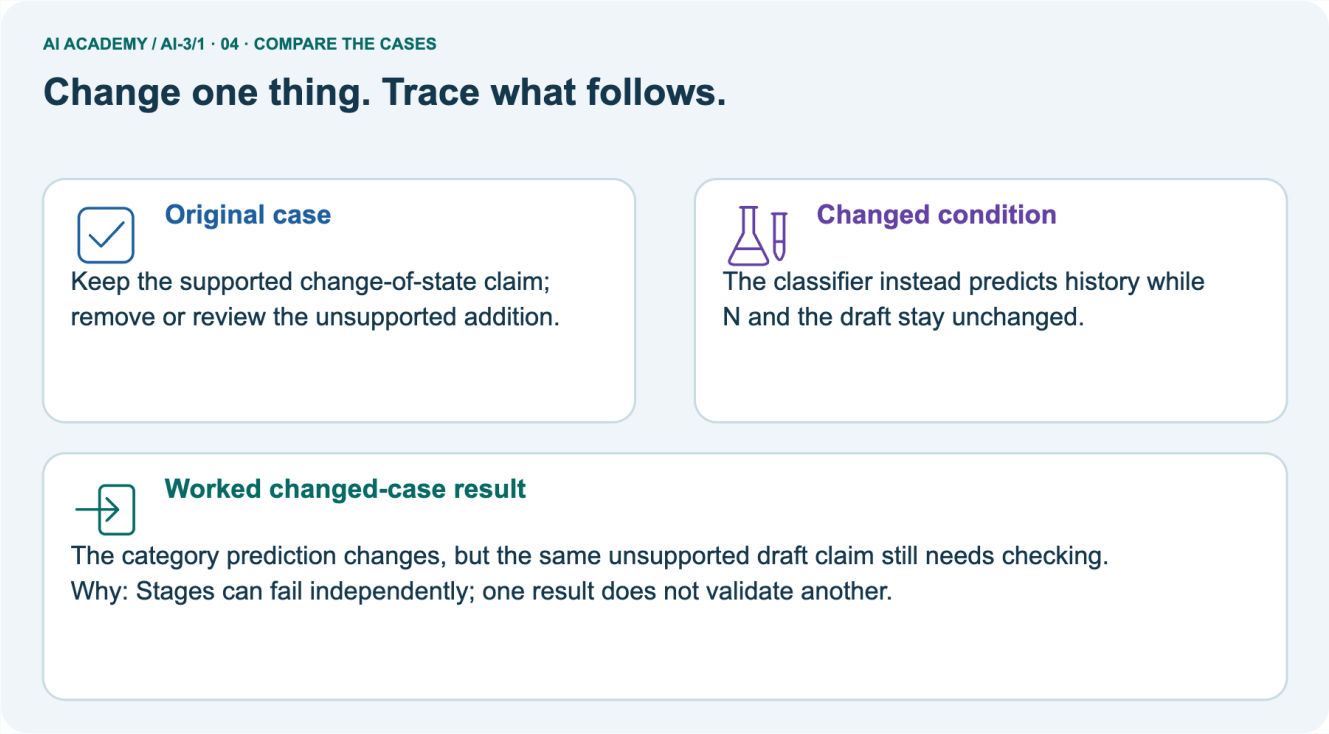
Follow the labelled picture with the matching explanation. Open the vector diagram

Keep the **evidence** roles separate. Ten corpus continuations | Count red 6, blue 3, green 1 | Distribution 0.6, 0.3, 0.1. Prefix: The marker is | Select a next token | An **output**, not an **observation**.

Current marker question | Check direct **evidence** | Colour claim needs support

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



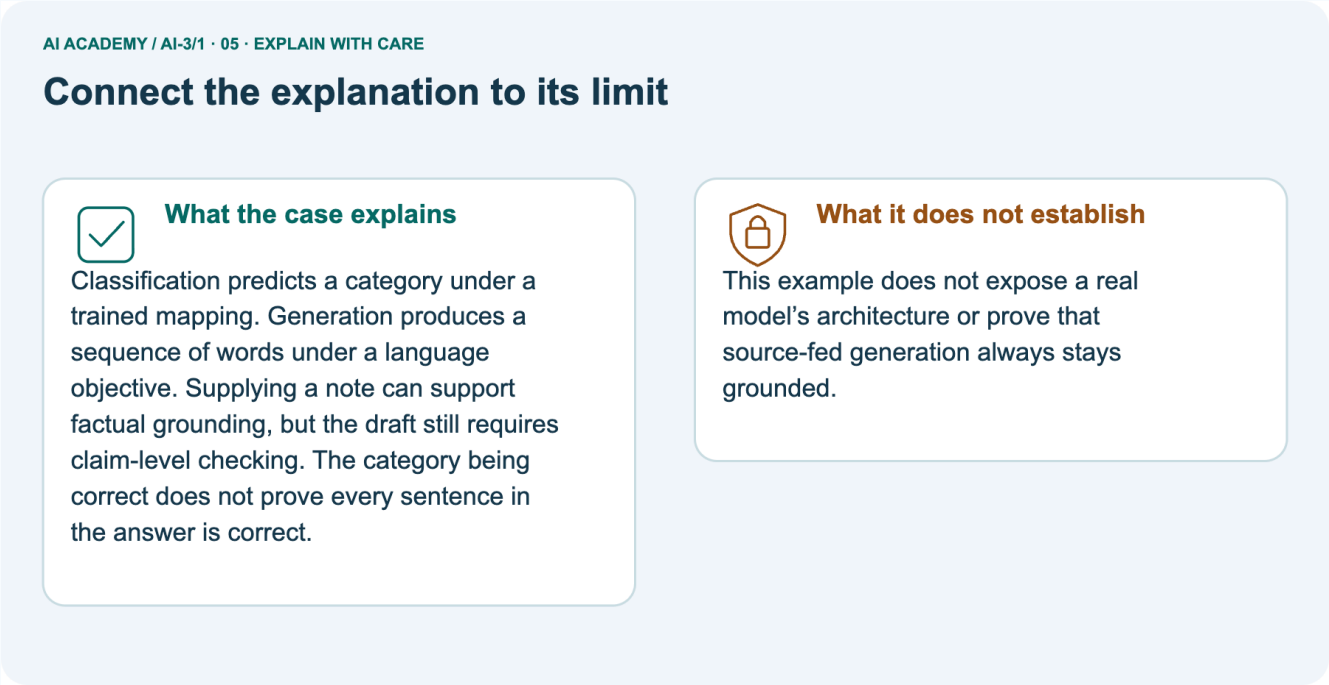
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Keep the supported change-of-state claim; remove or review the unsupported addition. Changed condition: The classifier instead predicts history while N and the draft stay unchanged.

Expected result: The category **prediction** changes, but the same unsupported draft claim still needs checking. Reason: Stages can fail independently; one result does not validate another.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



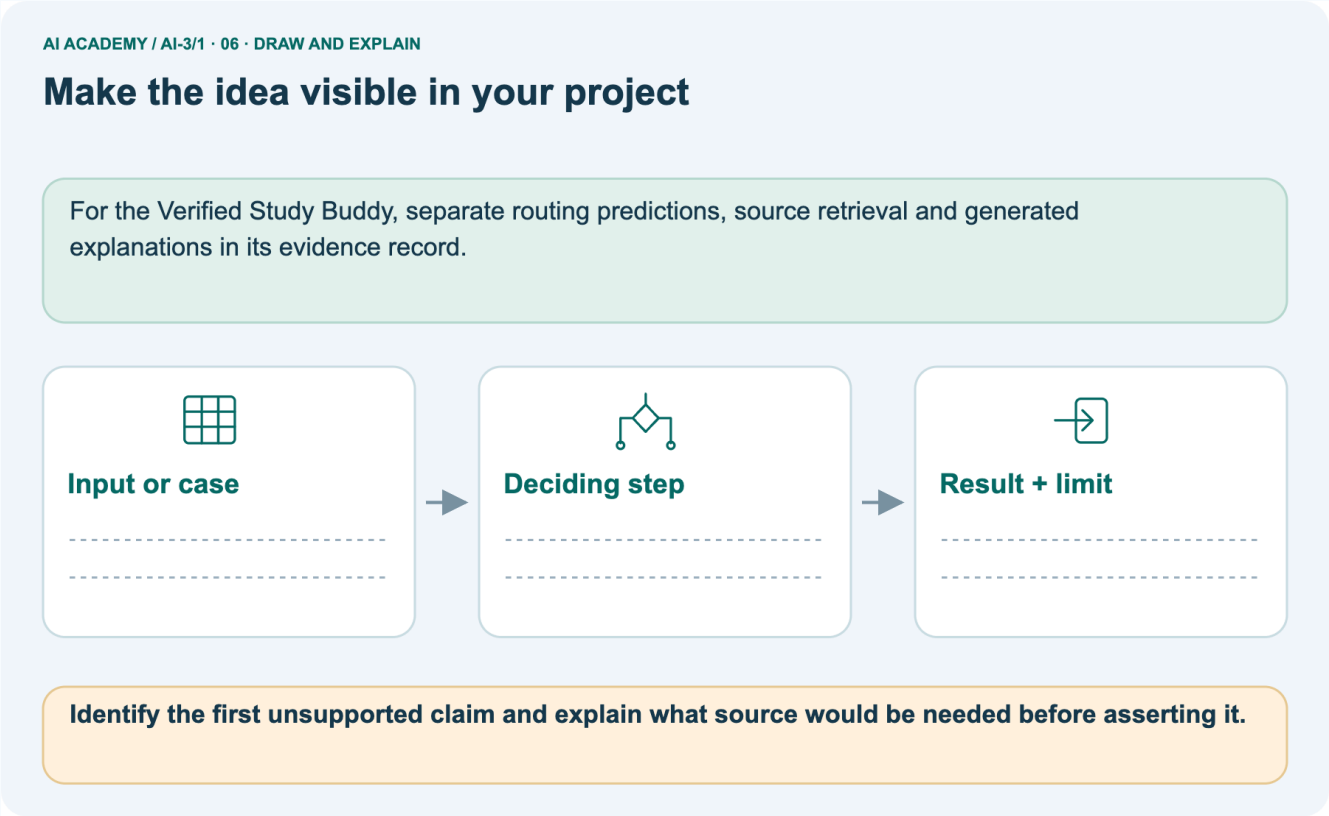
Follow the labelled picture with the matching explanation. Open the vector diagram

Classification predicts a category under a trained mapping. Generation produces a sequence of words under a language objective. Supplying a note can support factual grounding, but the draft still requires claim-level checking.

The category being correct does not prove every sentence in the answer is correct. Limit: This example does not expose a real model’s architecture or prove that source-fed generation always stays grounded.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Verified Study Buddy, separate routing **prediction**s, source retrieval and generated explanations in its **evidence** record. Draw your **input**, deciding step, result and limit.

Supported question: Identify the first unsupported claim and explain what source would be needed before asserting it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.